In [2]:
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```python
df = pd.read_excel('Superstore_USA.xlsx')
```

In [4]:
```python
df
```

Out[4]:

| | Row ID | Order Priority | Discount | Unit Price | Shipping Cost | Customer ID | Customer Name | Ship Mode | Cu Se |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 18606 | Not Specified | 0.01 | 2.88 | 0.50 | 2 | Janice Fletcher | Regular Air | Co |
| **1** | 20847 | High | 0.01 | 2.84 | 0.93 | 3 | Bonnie Potter | Express Air | Co |
| **2** | 23086 | Not Specified | 0.03 | 6.68 | 6.15 | 3 | Bonnie Potter | Express Air | Co |
| **3** | 23087 | Not Specified | 0.01 | 5.68 | 3.60 | 3 | Bonnie Potter | Regular Air | Co |
| **4** | 23088 | Not Specified | 0.00 | 205.99 | 2.50 | 3 | Bonnie Potter | Express Air | Co |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | |
| **9421** | 20275 | Critical | 0.06 | 35.89 | 14.72 | 3402 | Frederick Cole | Regular Air | Co |
| **9422** | 20276 | Critical | 0.00 | 3.34 | 7.49 | 3402 | Frederick Cole | Regular Air | Co |
| **9423** | 24491 | Not Specified | 0.08 | 550.98 | 45.70 | 3402 | Frederick Cole | Delivery Truck | Co |
| **9424** | 25914 | High | 0.10 | 105.98 | 13.99 | 3403 | Tammy Buckley | Express Air | Co |
| **9425** | 24492 | Not Specified | 0.09 | 7.78 | 2.50 | 3403 | Tammy Buckley | Express Air | Co |

9426 rows × 24 columns

In [6]:
```python
df.head(5)
```

Out[6]:

| | Row ID | Order Priority | Discount | Unit Price | Shipping Cost | Customer ID | Customer Name | Ship Mode | Custo Segm |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 18606 | Not Specified | 0.01 | 2.88 | 0.50 | 2 | Janice Fletcher | Regular Air | Corpo |
| **1** | 20847 | High | 0.01 | 2.84 | 0.93 | 3 | Bonnie Potter | Express Air | Corpo |
| **2** | 23086 | Not Specified | 0.03 | 6.68 | 6.15 | 3 | Bonnie Potter | Express Air | Corpo |
| **3** | 23087 | Not Specified | 0.01 | 5.68 | 3.60 | 3 | Bonnie Potter | Regular Air | Corpo |
| **4** | 23088 | Not Specified | 0.00 | 205.99 | 2.50 | 3 | Bonnie Potter | Express Air | Corpo |

5 rows × 24 columns

In [7]:
```
df.shape
```

Out[7]:
```
(9426, 24)
```

In [8]:
```
df.isnull()
```

Out[8]:

| | Row ID | Order Priority | Discount | Unit Price | Shipping Cost | Customer ID | Customer Name | Ship Mode | Custome Segme |
|---|---|---|---|---|---|---|---|---|---|
| **0** | False | False | False | False | False | False | False | False | Fal |
| **1** | False | False | False | False | False | False | False | False | Fal |
| **2** | False | False | False | False | False | False | False | False | Fal |
| **3** | False | False | False | False | False | False | False | False | Fal |
| **4** | False | False | False | False | False | False | False | False | Fal |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | |
| **9421** | False | False | False | False | False | False | False | False | Fal |
| **9422** | False | False | False | False | False | False | False | False | Fal |
| **9423** | False | False | False | False | False | False | False | False | Fal |
| **9424** | False | False | False | False | False | False | False | False | Fal |
| **9425** | False | False | False | False | False | False | False | False | Fal |

9426 rows × 24 columns

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
Row ID                    0
Order Priority            0
Discount                  0
Unit Price                0
Shipping Cost             0
Customer ID               0
Customer Name             0
Ship Mode                 0
Customer Segment          0
Product Category          0
Product Sub-Category      0
Product Container         0
Product Name              0
Product Base Margin       0
Region                    0
State or Province         0
City                      0
Postal Code               0
Order Date                0
Ship Date                 0
Profit                    0
Quantity ordered new      0
Sales                     0
Order ID                  0
dtype: int64
```

In [11]:
```
df['Product Base Margin'] = df['Product Base Margin'].fillna(df['Product Bas
```

In [14]:
```
df['Order Priority'].value_counts()
```

Out[14]:
```
Order Priority
High             1970
Low              1926
Not Specified    1881
Medium           1844
Critical         1804
Critical            1
Name: count, dtype: int64
```

In [62]:
```
# plt.figure(figsize=(6,4))
# sns.countplot(x="Order Priority", data=df)
# plt.title("Count of Order Priority")
# plt.savefig("Count of Order Priority.jpg")
# plt.show()
```

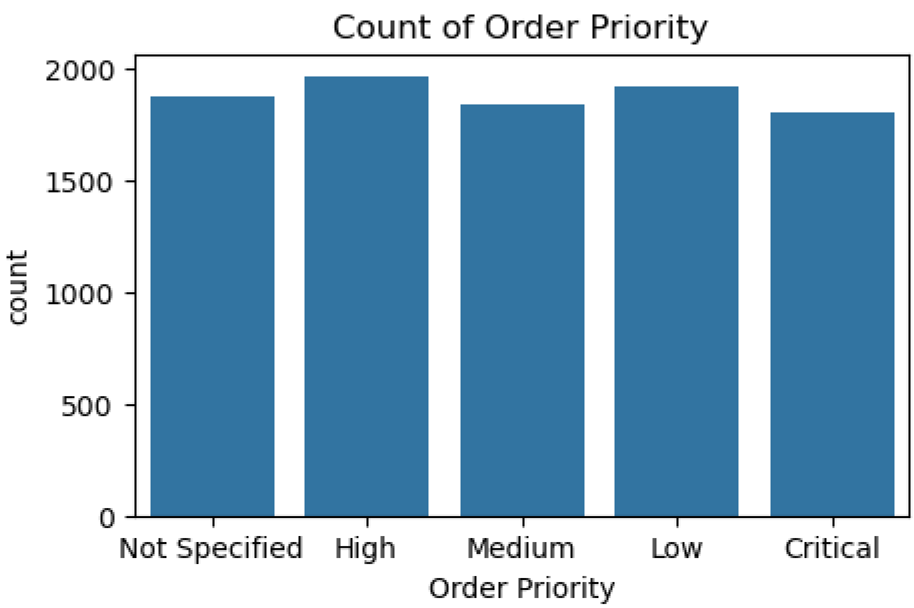
In [63]:
```
df['Order Priority'].unique()
```

Out[63]:
```
array(['Not Specified', 'High', 'Medium', 'Low', 'Critical', 'Critical '],
      dtype=object)
```

In [64]:
```
df['Order Priority'] = df['Order Priority'].replace('Critical ','Critical')
```

In [65]:
```
plt.figure(figsize=(5,3))
sns.countplot(x="Order Priority", data=df)
```

```
plt.title("Count of Order Priority")
plt.show()
```




In [78]:
```
df.columns
```

Out[78]:
```
Index(['Row ID', 'Order Priority', 'Discount', 'Unit Price', 'Shipping Cos
t',
       'Customer ID', 'Customer Name', 'Ship Mode', 'Customer Segment',
       'Product Category', 'Product Sub-Category', 'Product Container',
       'Product Name', 'Product Base Margin', 'Region', 'State or Provinc
e',
       'City', 'Postal Code', 'Order Date', 'Ship Date', 'Profit',
       'Quantity ordered new', 'Sales', 'Order ID', 'Order yer', 'Order yea
r'],
      dtype='object')
```

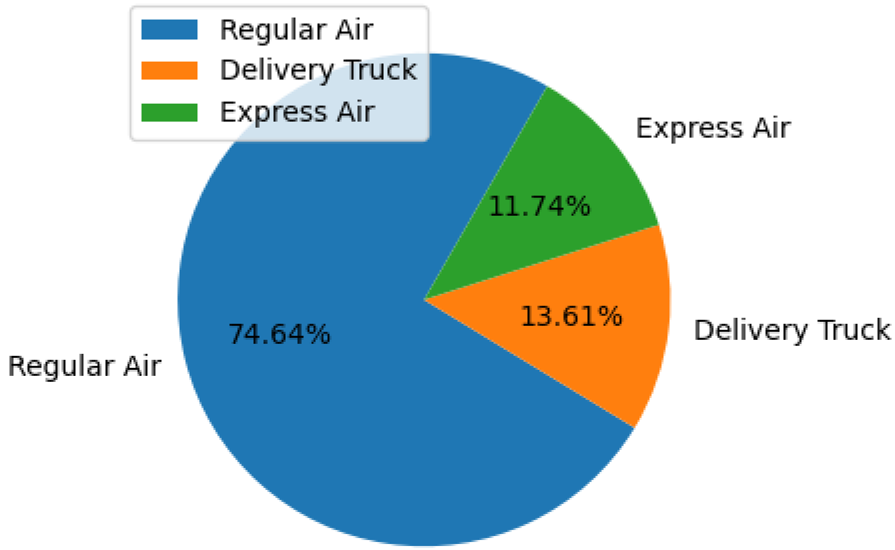
**Ship Mode**

In [67]:
```
df['Ship Mode'].value_counts()
```

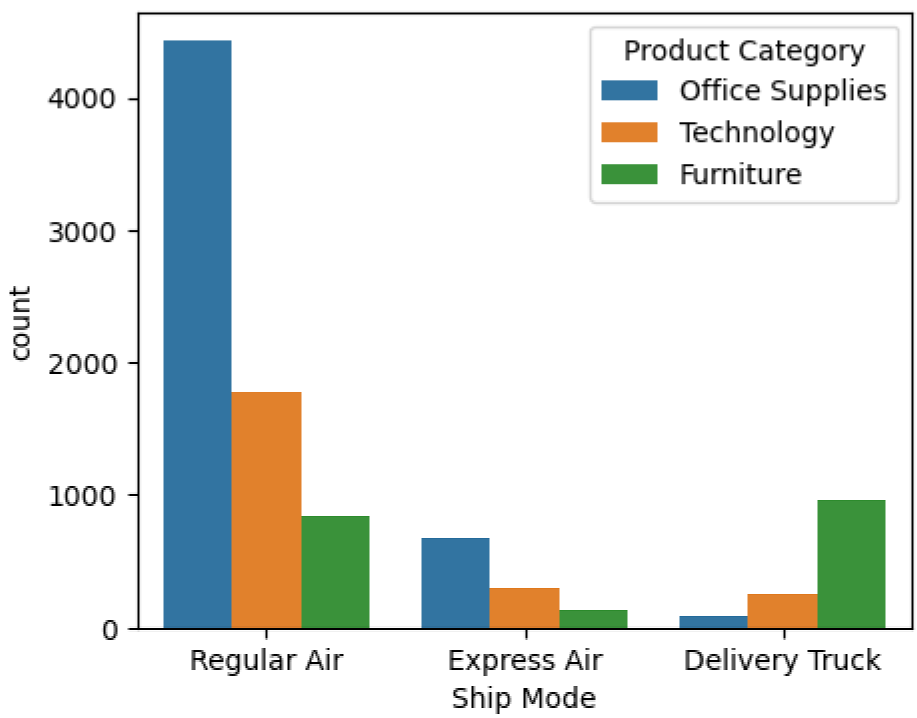
Out[67]:
```
Ship Mode
Regular Air       7036
Delivery Truck    1283
Express Air       1107
Name: count, dtype: int64
```

In [68]:
```
x = df['Ship Mode'].value_counts().index
y = df['Ship Mode'].value_counts().values
```

In [69]:
```
plt.figure(figsize=(5,4))
plt.pie(y, labels = x,startangle = 60,autopct="%0.2f%%")
plt.legend(loc=2)
plt.show()
```

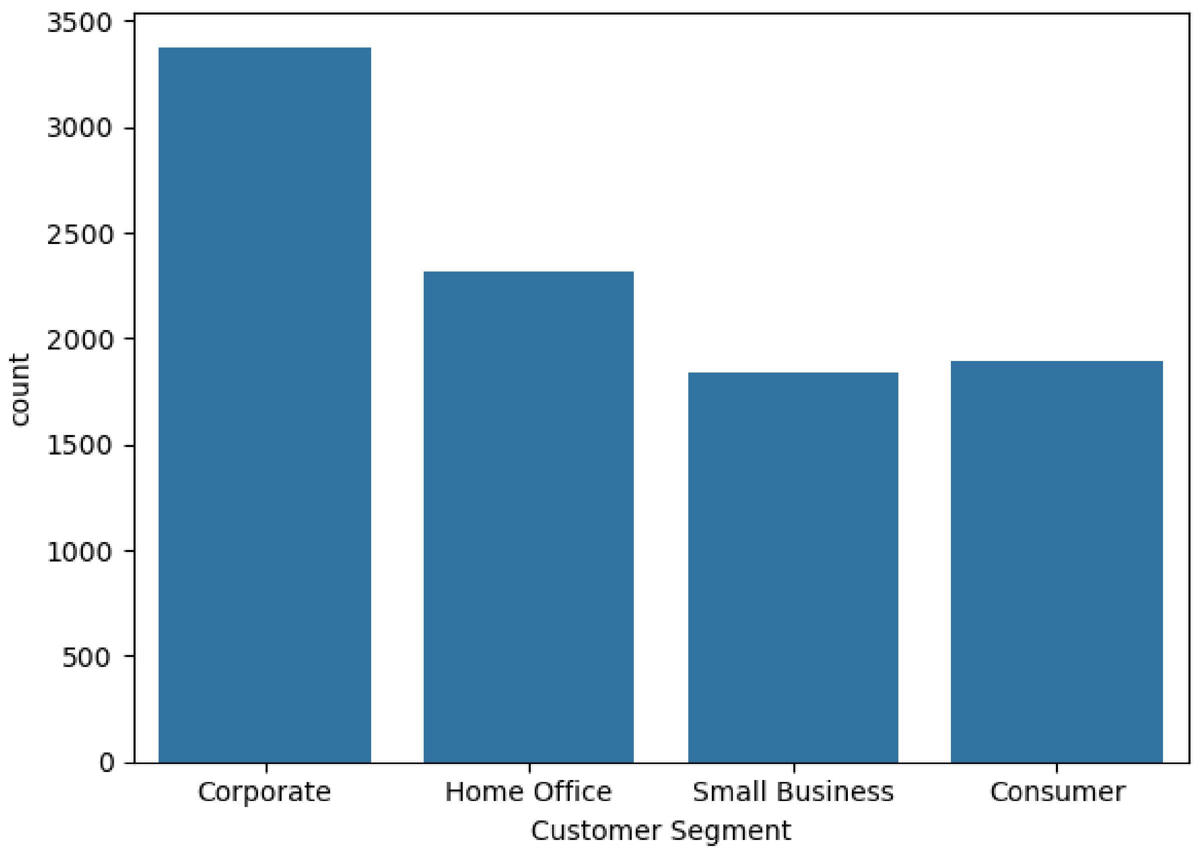



In [70]:
```
plt.figure(figsize=(5,4))
sns.countplot(x = "Ship Mode", data=df, hue="Product Category")
plt.show()
```




**Customer Segment**

In [71]:
```
plt.figure(figsize=(7,5))
sns.countplot(x="Customer Segment",data = df)
plt.show()
```




**Order year**

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9426 entries, 0 to 9425
Data columns (total 24 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Row ID                 9426 non-null   int64
 1   Order Priority         9426 non-null   object
 2   Discount               9426 non-null   float64
 3   Unit Price             9426 non-null   float64
 4   Shipping Cost          9426 non-null   float64
 5   Customer ID            9426 non-null   int64
 6   Customer Name          9426 non-null   object
 7   Ship Mode              9426 non-null   object
 8   Customer Segment       9426 non-null   object
 9   Product Category       9426 non-null   object
 10  Product Sub-Category   9426 non-null   object
 11  Product Container      9426 non-null   object
 12  Product Name           9426 non-null   object
 13  Product Base Margin    9354 non-null   float64
 14  Region                 9426 non-null   object
 15  State or Province      9426 non-null   object
 16  City                   9426 non-null   object
 17  Postal Code            9426 non-null   int64
 18  Order Date             9426 non-null   datetime64[ns]
 19  Ship Date              9426 non-null   datetime64[ns]
 20  Profit                 9426 non-null   float64
 21  Quantity ordered new   9426 non-null   int64
 22  Sales                  9426 non-null   float64
 23  Order ID               9426 non-null   int64
dtypes: datetime64[ns](2), float64(6), int64(5), object(11)
memory usage: 1.7+ MB
```

In [12]:
```
df['Order Date'].dt.year
```

Out[12]:
```
0       2012
1       2010
2       2011
3       2011
4       2011
        ...
9421    2013
9422    2013
9423    2013
9424    2010
9425    2013
Name: Order Date, Length: 9426, dtype: int32
```

In [17]:
```
df['Order year'] = df['Order Date'].dt.year
```

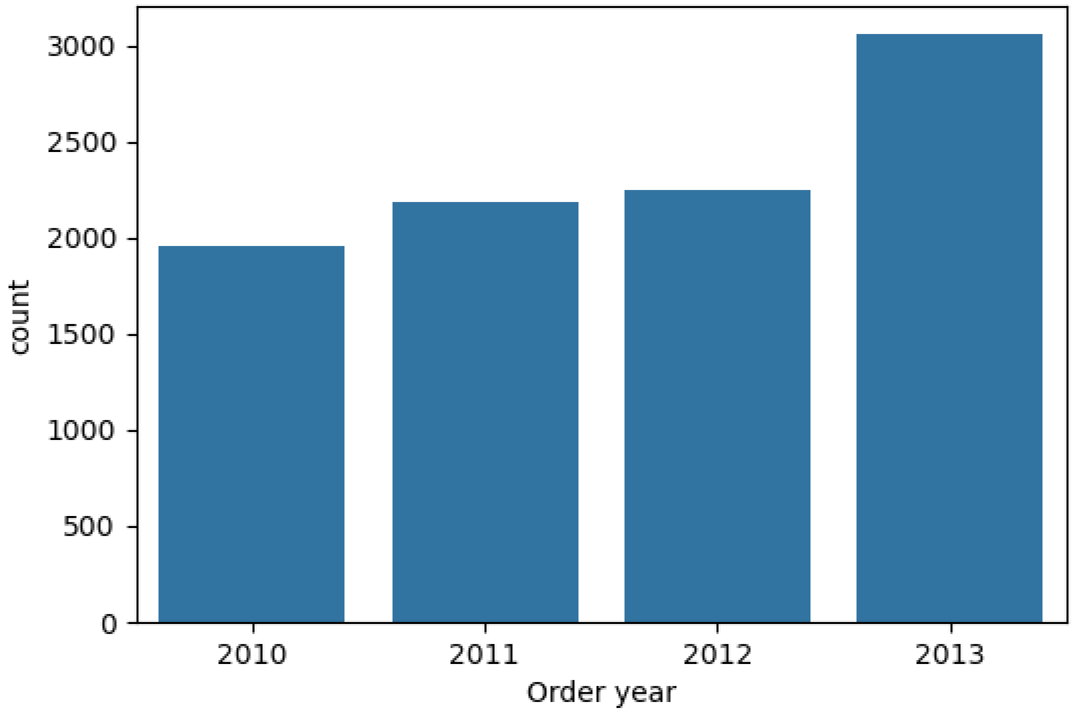
In [14]:
```
df.shape
```

Out[14]:
```
(9426, 25)
```

In [19]:
```
df['Order year'].value_counts()
```

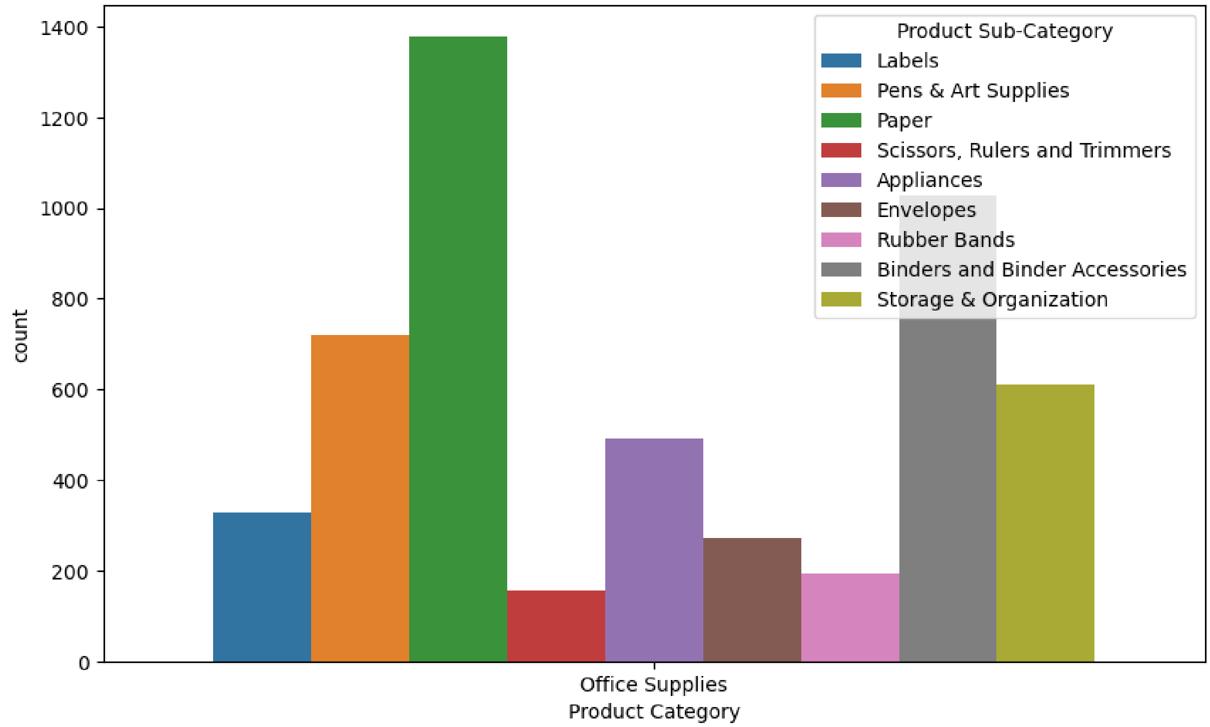
Out[19]:
```
Order year
2013    3054
2012    2241
2011    2179
2010    1952
Name: count, dtype: int64
```

In [16]:
```
plt.figure(figsize=(6,4))
sns.countplot(x="Order year",data = df)
plt.show()
```

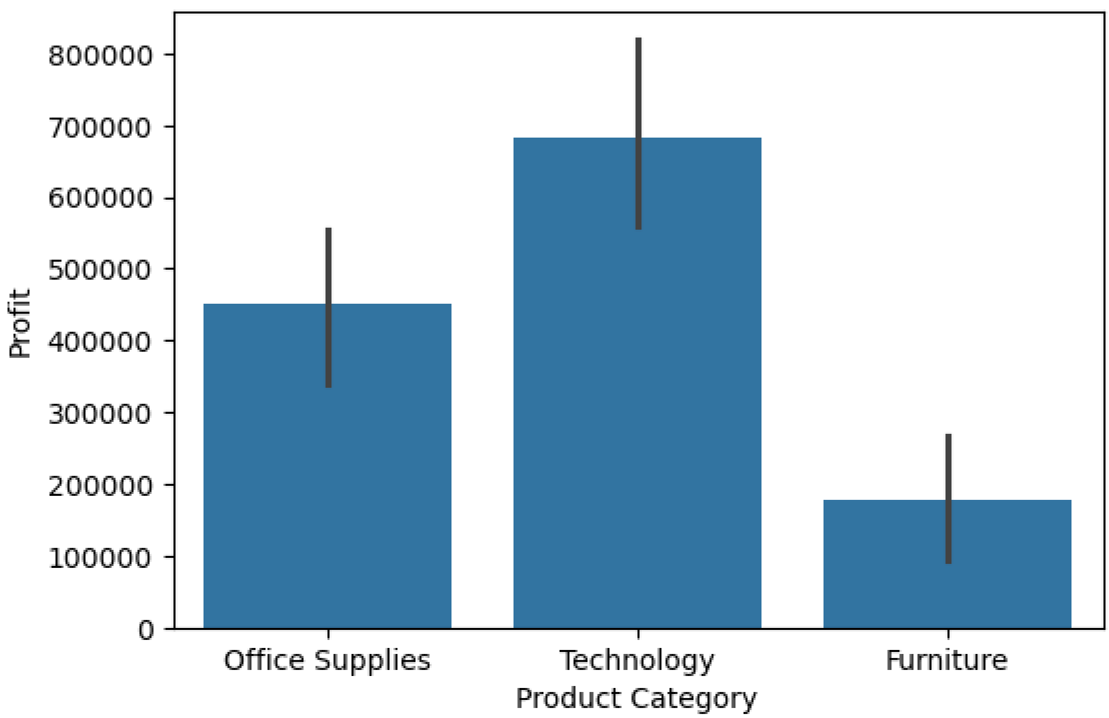



**Product Category**

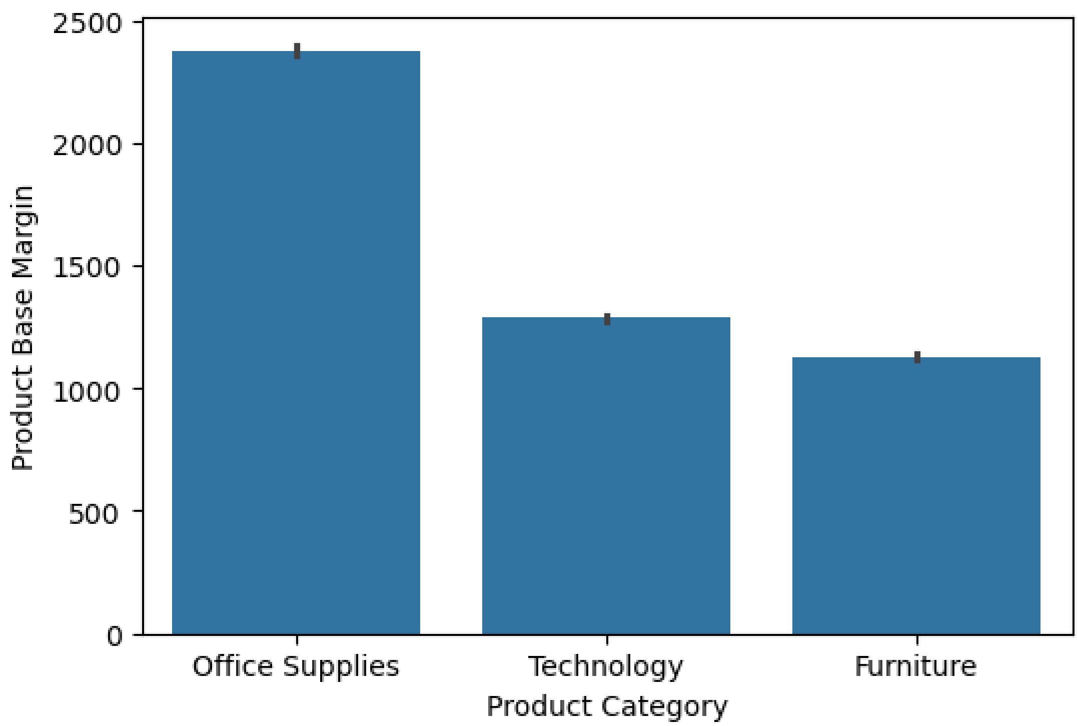
In [10]:
```
plt.figure(figsize=(10,6))
sns.countplot(x="Product Category",data=df[df["Product Category"] == "Office
plt.show()
```




In [11]:
```
plt.figure(figsize=(6,4))
sns.barplot(x="Product Category",y="Profit",data = df, estimator='sum')
plt.show()
```




In [77]:
```
plt.figure(figsize=(6,4))
sns.barplot(x="Product Category",y="Product Base Margin",data = df, estimato
plt.show()
```

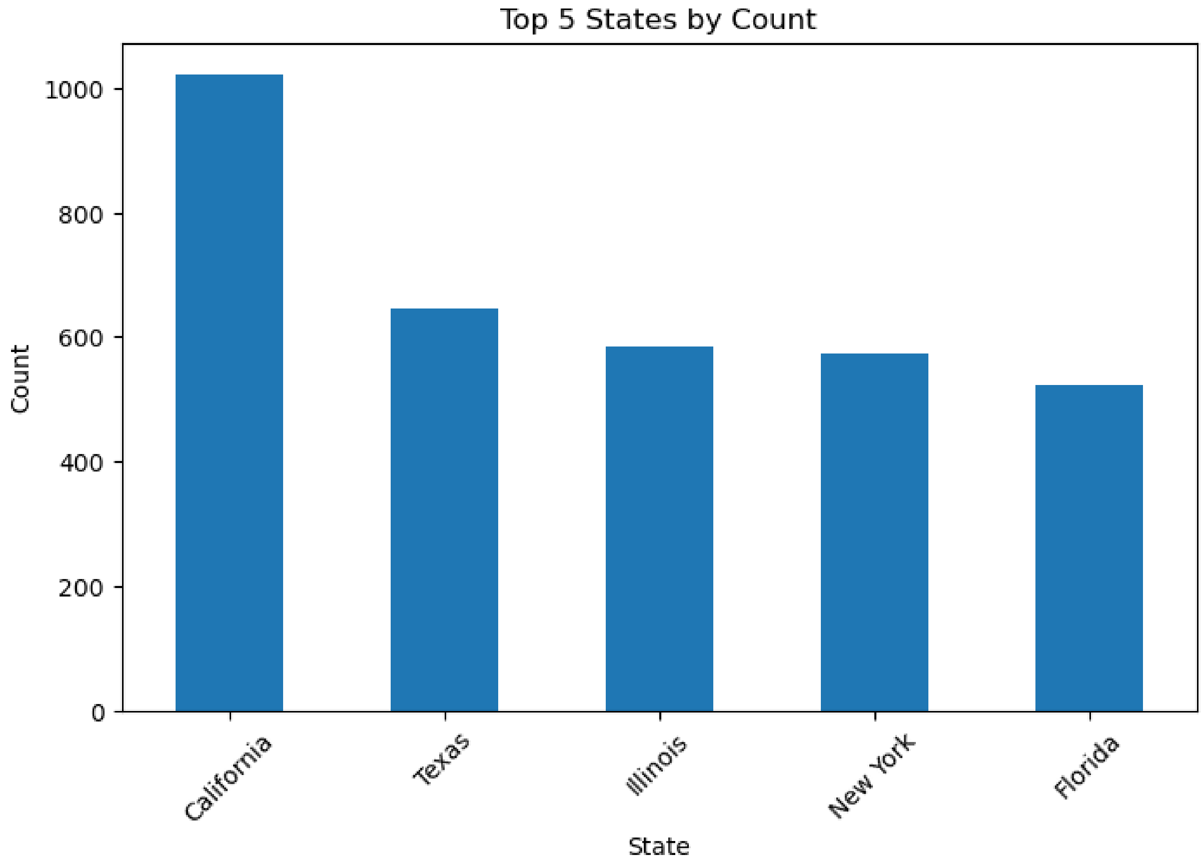



**State or Province**

In [83]:
```
df['State or Province'].value_counts()[:5]
```

Out[83]:
```
State or Province
California    1021
Texas          646
Illinois       584
New York       574
Florida        522
Name: count, dtype: int64
```

In [86]:
```
plt.figure(figsize=(8,5))
df['State or Province'].value_counts().head(5).plot(kind='bar')
plt.xlabel("State")
plt.ylabel("Count")
plt.title("Top 5 States by Count")
plt.xticks(rotation=45)
plt.show()
```


Top 5 States by Count
Count
1000
800
600
400
200
0
California
Texas
Illinois
New York
Florida
State

In [ ]: